

Aleksei Gregory Sorokin

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Research Scientific Machine Learning, Gaussian Processes, Quasi-Monte Carlo, Probabilistic Numerics

Programming Python (PyTorch, GPyTorch, Pandas, Matplotlib), Julia, C, MATLAB, R, SQL, Wolfram

Tools AWS (SageMaker, EC2), GitHub (general, actions, pages), L^AT_EX, Docker

Education

2026.01 - 2028.05 **Postdoc in Statistics.** University of Chicago. Advisors *Yuehaw Khoo and Lek-Heng Lim*.

2021.08 - 2025.12 **PhD in Applied Math.** Illinois Institute of Technology (IIT). GPA 3.89/4. Advisor *Fred J Hickernell*.

2017.08 - 2021.05 **Master of Data Science.** IIT. Summa Cum Laude. GPA 3.94/4.

2017.08 - 2021.05 **B.S. in Applied Math, Minor in Computer Science.** IIT. Summa Cum Laude. GPA 3.94/4.

Experiences

2025.01 - 2025.12 **DOE SCGSR Fellow in Applied Math at Sandia National Laboratory** in Livermore, CA. I produced scientific ML models for machine-precision solutions to nonlinear PDEs [5]. I developed scalable multifidelity Gaussian processes regression models and open-source software implementations [9, 10, 4].

2024.05 - 2024.08 **Scientific Machine Learning Researcher at FM (Factory Mutual Insurance Company).** I deployed scientific ML models, including PINNs DeepONets, to accelerate CFD fire dynamics simulations [12].

2023.05 - 2023.08 **Graduate Intern at Los Alamos National Laboratory.** I modeled multifidelity solutions to PDE with random coefficients using efficient and error aware Gaussian processes regression models [13].

2022.05 - 2022.08 **Givens Associate Intern at Argonne National Laboratory.** I derived error bounds and a sequential sampling method for efficiently estimating failure probabilities with probabilistic models [3].

2021.05 - 2021.08 **ML Engineer Intern at SigOpt, an Intel Company.** In a six-person ML team, I contributed production code for meta-learning model-aware hyperparameter tuning via Bayesian optimization [16].

2022.09 - 2022.11 **Participant in Argonne National Laboratory's Course on AI Driven Science on Supercomputers.** Key topics included handling large scale data pipelines and parallel training for neural networks.

2018.05 - 2019.08 **Instructor for the STARS Computing Corps' Computer Discover Program.** I taught and developed curriculum for middle school and high school girls to learn programmatic thinking in Python.

2021.08 - 2025.01 **Teaching Assistant at IIT.** I led reviews for PhD qualifying exams in analysis and computational math.

Open-Source Software

QMCPy Quasi-Monte Carlo Python Software (qmcsoftware.github.io/QMCSoftware). I led dozens of col-laborators across academia and industry to develop QMC sequence generators, automatic variable transformations, adaptive error estimators, and diverse use cases [2, 7, 8, 17, 18, 19, 20, 11, 1].

FastGPs Scalable Gaussian Processes in Python (alegresor.github.io/fastgps). This supports GPU scaling, batched inference, hyperparameter optimization, multifidelity GPs, and efficient Bayesian cubature. FastGPs is the first package to implement GPs which require only $\mathcal{O}(n)$ storage and $\mathcal{O}(n \log n)$ computations compared to the typical $\mathcal{O}(n^2)$ storage and $\mathcal{O}(n^3)$ computations requirements [10, 9, 4].

QMCGenerators.jl Randomized Quasi-Monte Carlo Sequences in Julia (alegresor.github.io/QMCGenerators.jl).

QMCToolsCL Randomized Quasi-Monte Carlo Sequences in C/OpenCL (qmcsoftware.github.io/QMCToolsCL/).

TorchOrthoPolys Orthogonal Polynomials in PyTorch (alegresor.github.io/TorchOrthoPolys/) with GPU support.

Awards

2025.01 - 2025.12 **DOE SCGSR Fellow in Applied Math**, Sandia National Laboratory, Livermore California.

2025.01 **Karl Menger Student Award for Exceptional Scholarship (Graduate)**, IIT.

2024.01 **College of Computing Excellence in Dissertation Research**, IIT.

2024 **Teaching Assistant Award**, IIT.

2023.08 **Outstanding Math Poster**, Los Alamos National Laboratory.

2017.08 - 2025.05 **Deans List Member**, IIT, every semester.

References

PhD Advisor **Fred J. Hickernell** (hickernell@iit.edu) Vice Provost for Research and Professor of Applied Math, IIT.

Mentor **Nicolas W. Hengartner** (nickh@lanl.gov) Senior Scientist, Los Alamos National Lab.

Mentor **Michael J. McCourt** (mikemccourt1234@gmail.com) CTO and Co-Founder at Distributional.

Mentor **Pieterjan M. Robbe** (pmrobbe@sandia.gov) Senior Member of Technical Staff, Sandia National Lab.

- [1] Aadit Jain, Fred J. Hickernell, Art B. Owen, and Aleksei G. Sorokin. "Empirical Bernstein and betting confidence intervals for randomized quasi-Monte Carlo". In: *Information and Inference: A Journal of the IMA* 15.1 (Mar. 2026), iaag003. ISSN: 2049-8772. DOI: 10.1093/imaiai/iaag003. eprint: <https://academic.oup.com/imaiai/article-pdf/15/1/iaag003/67259522/iaag003.pdf>. URL: <https://doi.org/10.1093/imaiai/iaag003>.
- [2] Aleksei G. Sorokin, Fred J. Hickernell, Sou-Cheng T. Choi, Jagadeeswaran Rathinavel, Pieterjan Robbe, and Aadit Jain. "QMCPy: a Python framework for (quasi-)Monte Carlo algorithms". In: *Journal of Open Source Software* 11.117 (2026), p. 9705. DOI: 10.21105/joss.09705. URL: <https://doi.org/10.21105/joss.09705>.
- [3] Aleksei G. Sorokin and Vishwas Rao. "Credible intervals for probability of failure with Gaussian processes". In: (2026). DOI: 10.48550/arXiv.2311.07733.
- [4] Aleksei G. Sorokin, Pieterjan Robbe, and Fred J. Hickernell. *Fast multitask Gaussian process regression*. 2026. DOI: 10.48550/arXiv.2603.16014.
- [5] Aras Bacho, Aleksei G. Sorokin, Xianjin Yang, Théo Bourdais, Edoardo Calvello, Matthieu Darcy, Alexander Hsu, Bamdad Hosseini, and Houman Owhadi. "Operator learning at machine precision". In: (2025). DOI: 10.48550/arXiv.2511.19980.
- [6] Eda Gjergo, Zhi-Yu Zhang, Pavel Kroupa, Aleksei G. Sorokin, Zhiqiang Yan, Ziyi Guo, Tereza Jerabkova, Akram Hasani Zonoozi, and Hosein Haghi. "Massive Star Formation at Supersolar Metallicities: Constraints on the Initial Mass Function". In: *Research in Astronomy and Astrophysics* 26.2 (Jan. 2025), p. 025003. DOI: 10.1088/1674-4527/ae1f79.
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